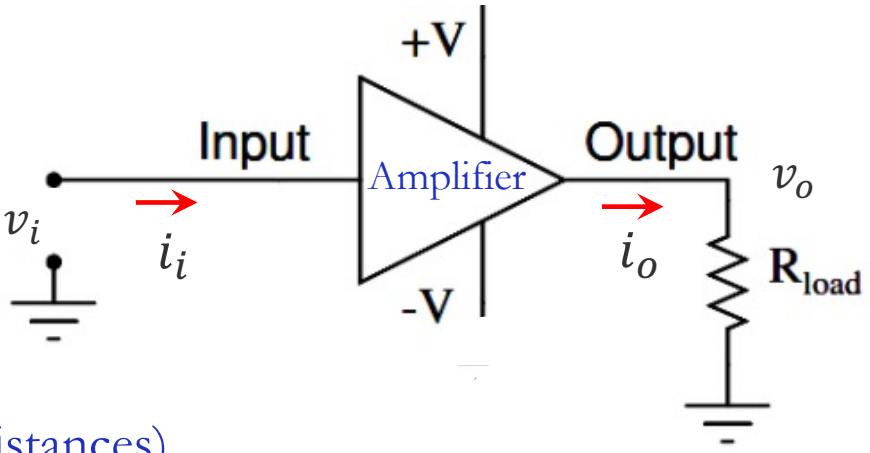
Field Effect Transistors

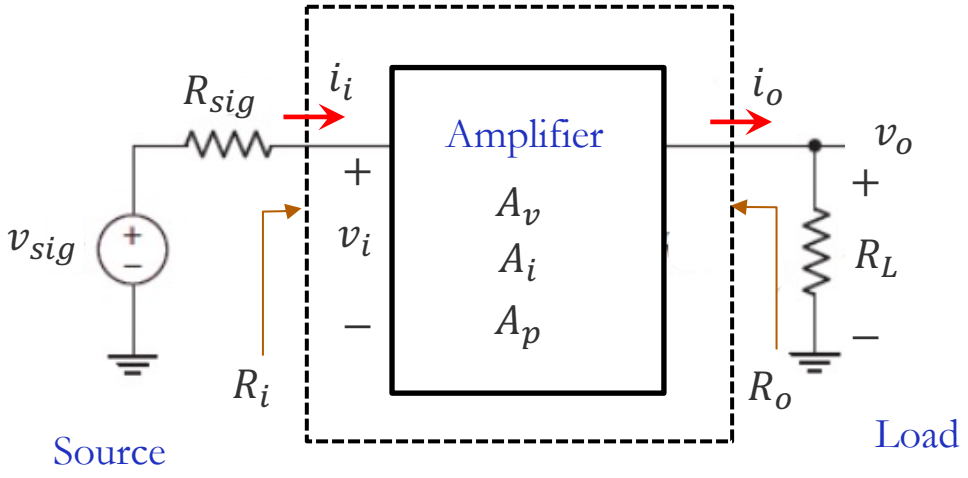
- The MOSFET as an Amplifier.
- Small Signal Analysis.

Amplifiers: Introduction

Basic requirements of any amplifier are:

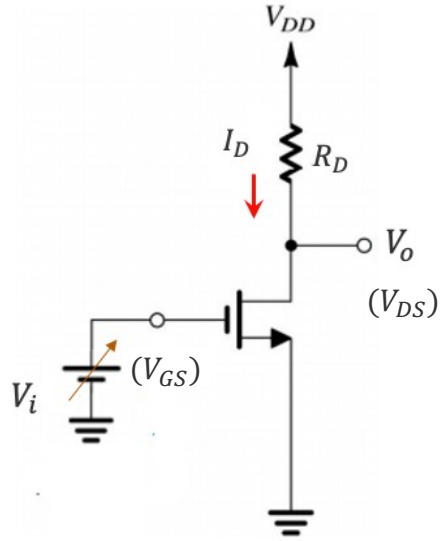
1) Gain (voltage, current, power)
2) Linear operation.
3) Large swing (dynamic range).
4) Suitable coupling (input and output resistances).

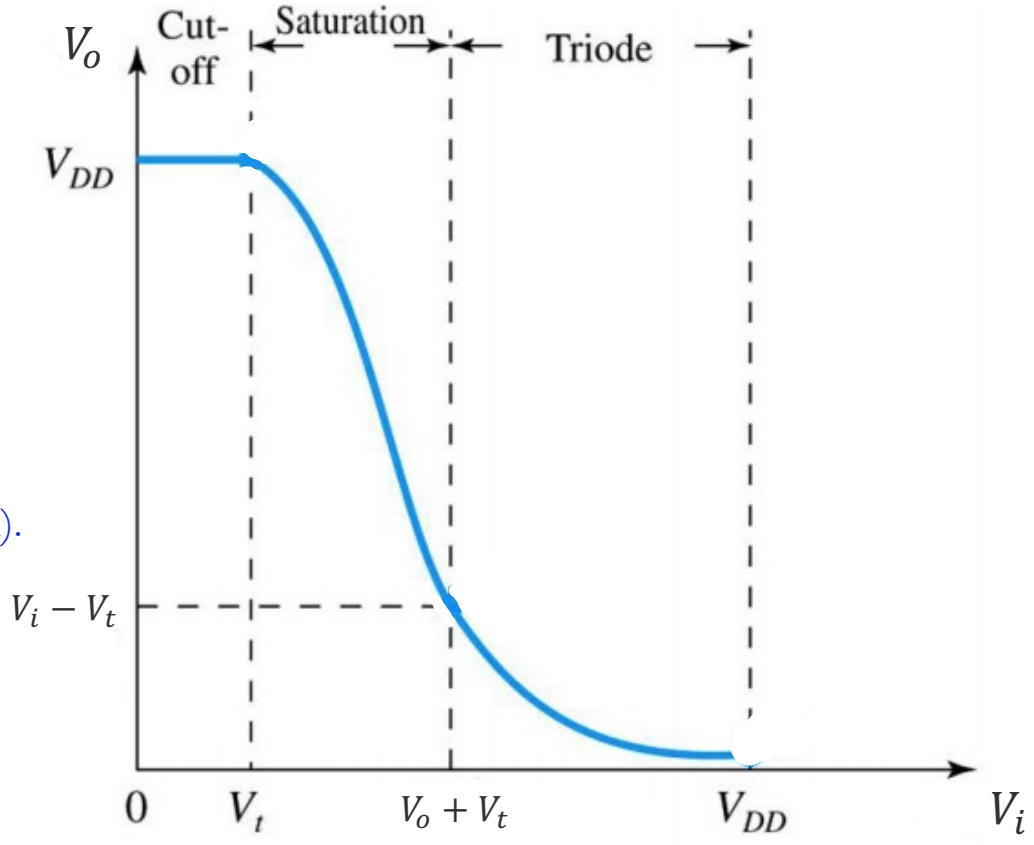




MOSFET Transfer Function

Consider the following DC transistor circuit.





1) V_i (small) $< V_t$: Cutoff $I_D = 0, V_o = V_{DD}$ (high).

2) $V_i > V_t : I_D \neq 0, \quad V_o = V_{DD} - I_D R_D$

V_o is still high but decreasing

Saturation

3) V_i (high) $\gg V_t$: $I_D \neq 0, \quad V_o = V_{DD} - I_D R_D$

V_o has decreased too much

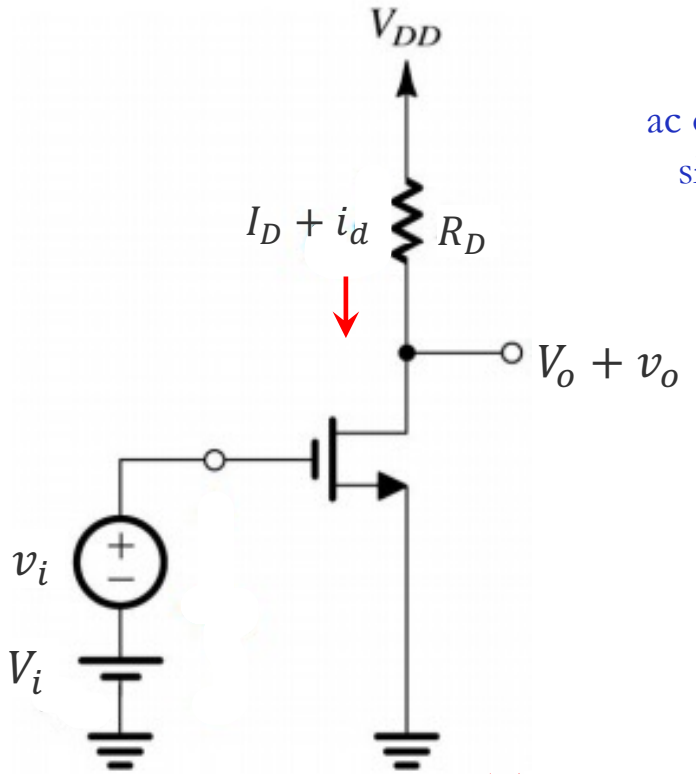
Triode

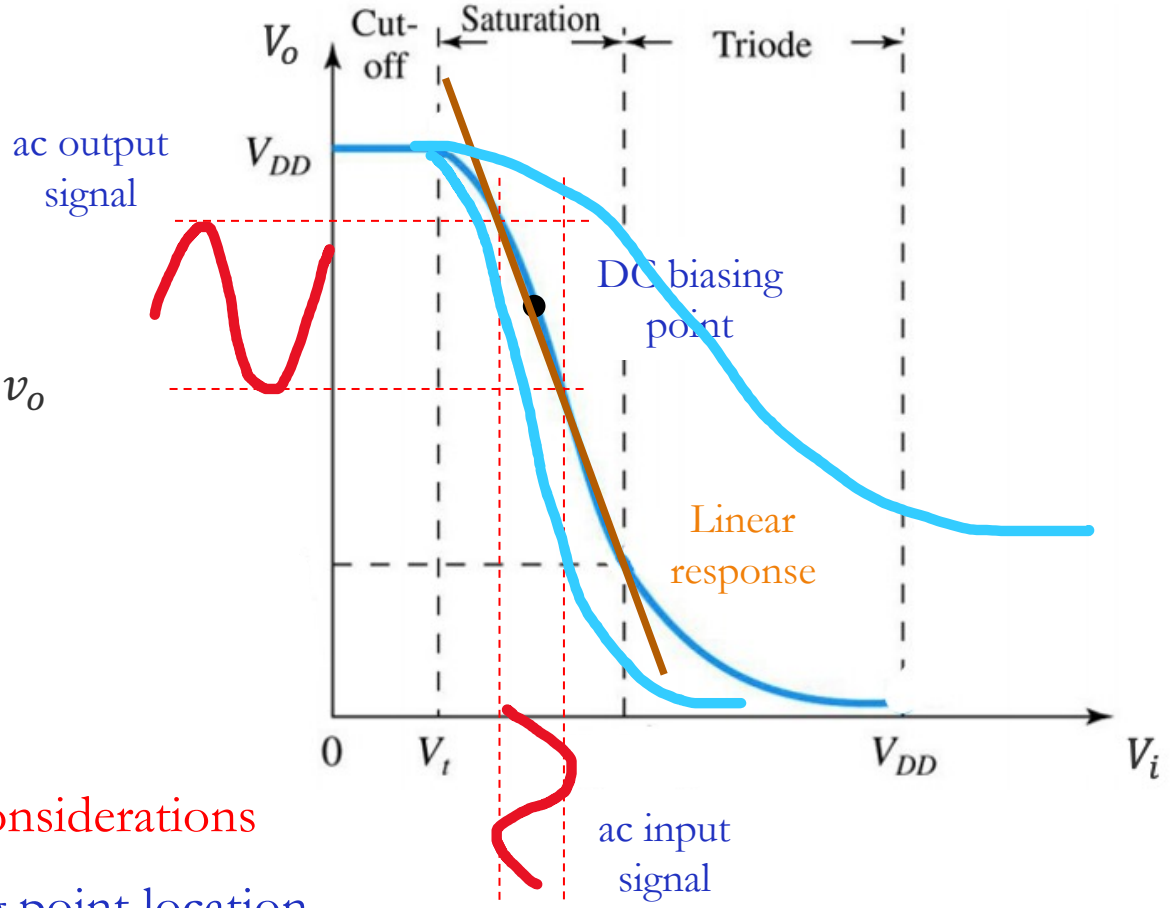
Which mode is suitable for amplification?

Remember the basic requirements.

MOSFET Transfer Function

Now an ac source is added at the input.





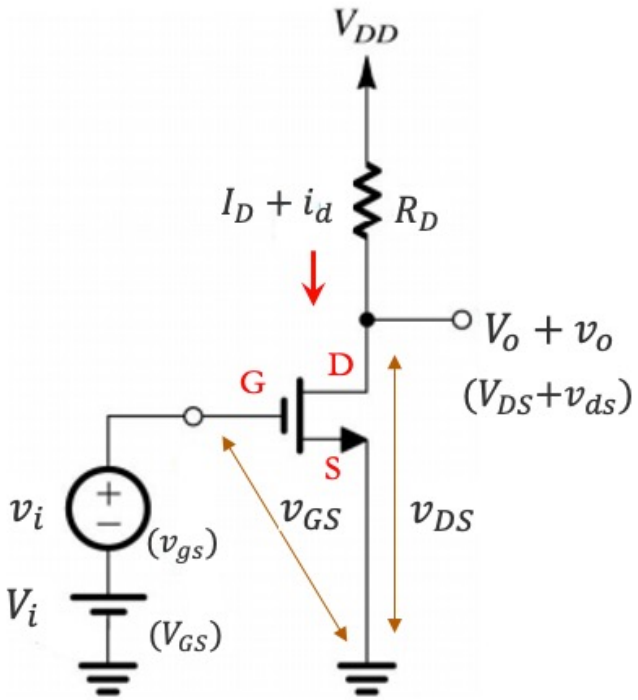
Two considerations

1) Biasing point location

2) How large should the ac signal be?

The MOSFET as an Amplifiers

Reconsider the previous circuit. Let's write the current equation.



Note the symbol convention
DC: all capital
ac: all small
Total: mixed

The transistor is in saturation.

$$i_D = k(v_{GS} - V_t)^2 \quad \Rightarrow \quad i_D = k\left(V_{GS} + v_{gs} - V_t\right)^2$$

$$i_D = \underbrace{k(V_{GS} - V_t)^2}_{I_D \text{ (DC part)}} + \underbrace{2k(V_{GS} - V_t)v_{gs} + k v_{gs}^2}_{i_d \text{ (ac part)}}$$

For a small input signal (small-signal approximation)

$$i_D = I_D + i_d = k(V_{GS} - V_t)^2 + 2k(V_{GS} - V_t)v_{gs}$$

$$i_d = 2k(V_{GS} - V_t)\, v_{gs} = g_m\, v_{gs}$$ i_d is linear with ac input

$$v_O = V_o + v_o = V_{DD} - (I_D + i_d)R_D = \underbrace{(V_{DD} - I_D R_D)}_{\text{(DC part)}} \underbrace{- i_d R_D}_{\text{(ac part)}}$$

$$v_o = -i_d\, R_D = -g_m R_D\, v_{gs}$$ v_o is linear with ac input

$$i_d = g_m\, v_i$$

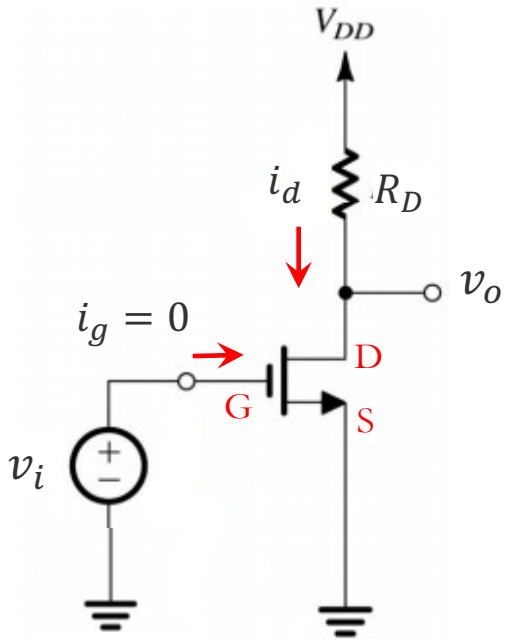
$$v_o = -g_m R_D\, v_i$$

$$g_m = 2k(V_{GS} - V_t) = 2\sqrt{k}\sqrt{I_D}$$

Transconductance (A/V)

The MOSFET as an Amplifiers

Important conclusions regarding the operation of the MOSFET as an amplifier.



1) The amplifier voltage gain is given as v_o/v_i.

For this circuit $\dfrac{v_o}{v_i} = -g_m R_D$

2) The gain is a function of the DC status (bias).

3) The transconductance g_m is a function of the DC status (bias) and transistor dimensions.

4) The transistor's small-signal current and voltage relations can be converted to a "circuit model".

$i_d = g_m\, v_i$

$v_o = -g_m R_D\, v_i$

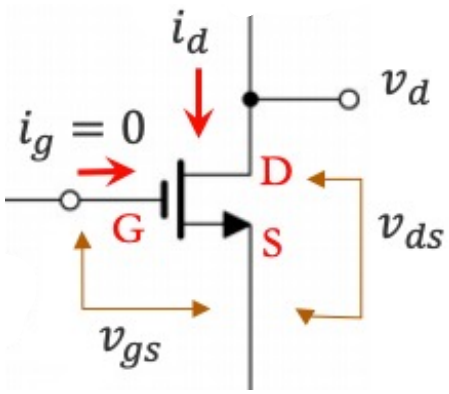
$g_m = 2k(V_{GS} - V_t) = 2\sqrt{k}\sqrt{I_D}$

$i_g = 0$ ➡ Open circuit

$i_d = g_m\, v_{gs}$ ➡ Controlled current source

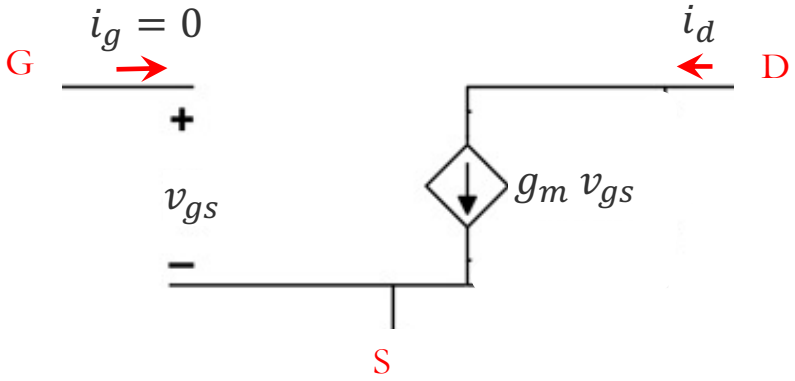
$\lambda \neq 0$ ➡ Output resistance

MOSFET Small-signal Circuit Model

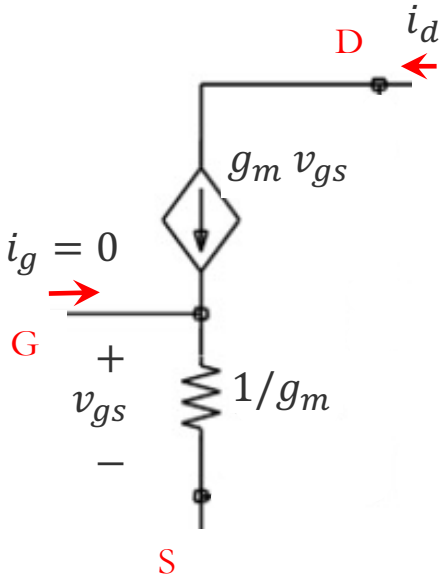


The transistor's small-signal current and voltage relations can be converted to a "circuit model".

$i_g = 0$	➡	Open circuit
$i_d = g_m\, v_i$	➡	Controlled current source
$\lambda \neq 0$	➡	Output resistance $r_o = V_A / I_D$



The NMOS π Model



The NMOS T Model

Small-signal Analysis

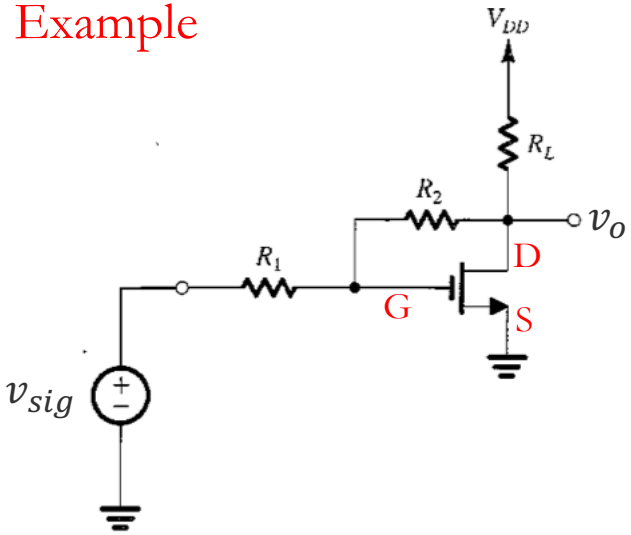
Procedure

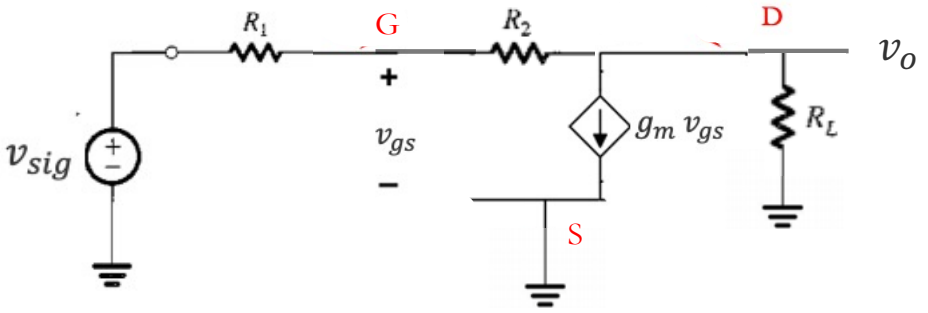
1) Perform DC analysis of the circuit. Find I_D and hence g_m and r_o.

2) Produce the small-signal equivalent circuit using the pi or T model of the transistor.

3) Analyze the circuit using circuit equations (KVL, KCL, etc.) to find important amplifier parameters (gain, input/output resistances, etc.)

Example

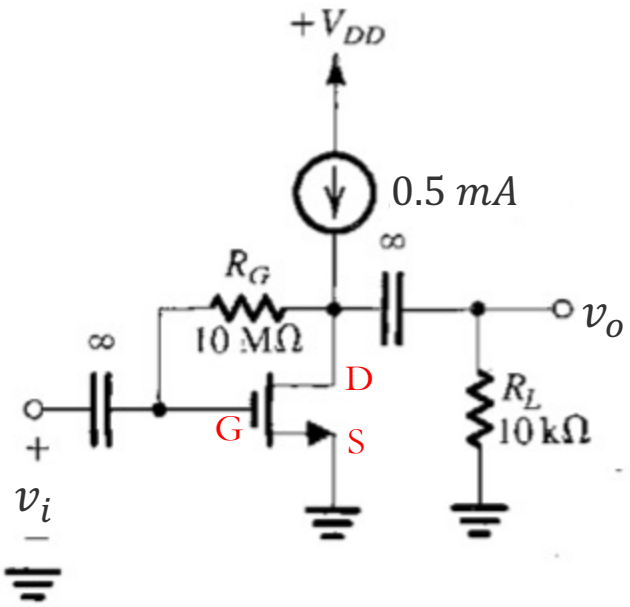




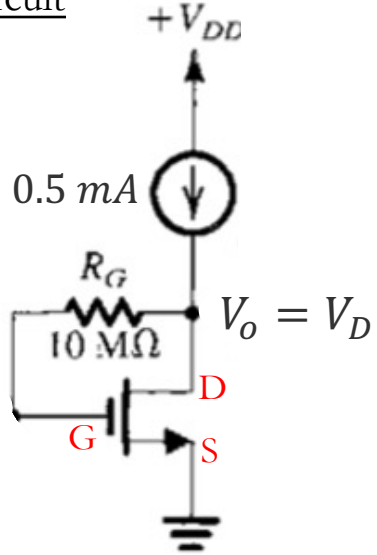
Small-signal equivalent circuit

Example

Find the voltage gain, input and output resistances of the amplifier circuit shown below. Take $V_t = 0.5\ V, V_A = 50\ V$ and $k = 2\ mA/V^2$.



DC Circuit



$I_D = 0.5\ mA$

$g_m = 2\sqrt{k}\sqrt{I_D} = 2\ mA/V$

$r_o = \dfrac{V_A}{I_D} = 100\ k\Omega$

DC Analysis

Transistor is in saturation $(V_D = V_G)$

$I_D = k(V_{GS} - V_t)^2$

$0.5\ m = 2\ m\ (V_D - 0.5)^2$

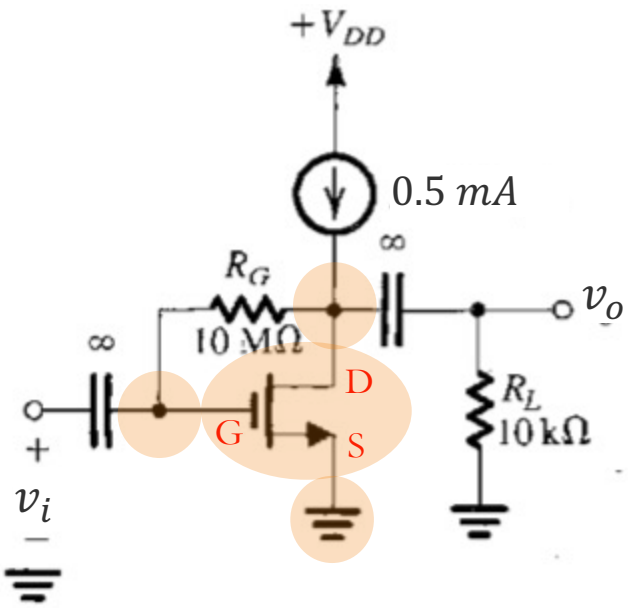
$V_D = +1\ V$

$V_o = +1\ V$

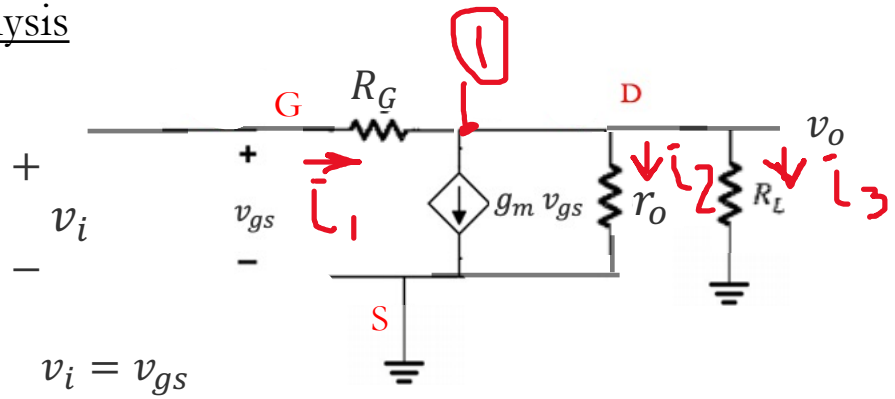
Capacitors protect the DC conditions and clean the ac output.
Coupling and bypass.

Example

Find the voltage gain, input and output resistances of the amplifier circuit shown below. Take $V_t = 0.5\,V, V_A = 50\,V$ and $k = 2\,mA/V^2$.



AC Analysis



$v_i = v_{gs}$

KCL at node 1 $\quad i_1 = g_m v_{gs} + i_2 + i_3$

$$\frac{v_i - v_o}{R_G} = g_m v_i + \frac{v_o}{r_o} + \frac{v_o}{R_L}$$

$$-v_o\left(\frac{1}{R_G} + \frac{1}{r_o} + \frac{1}{R_L}\right) = v_i\left(g_m - \frac{1}{R_G}\right)$$

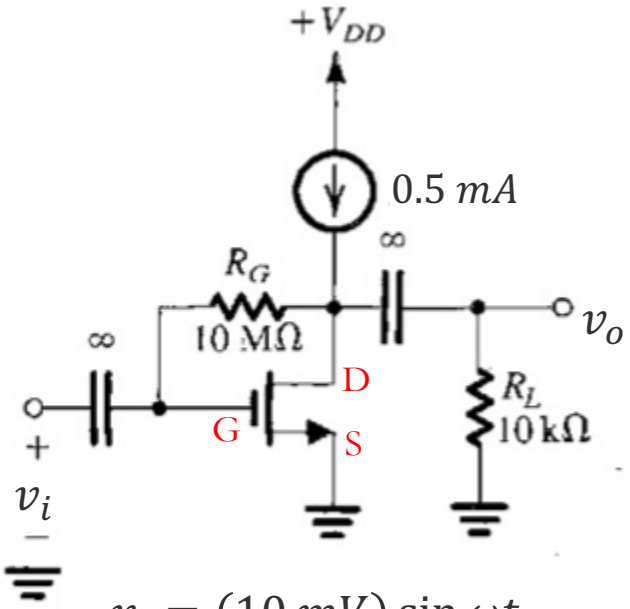
If we ignore r_o,

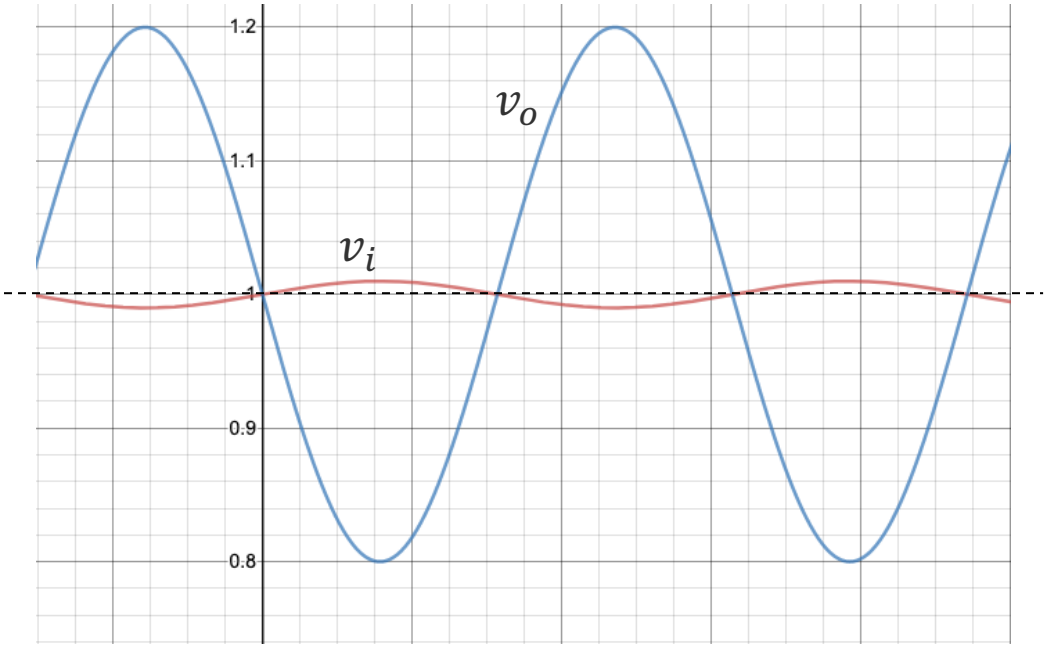
$$A_v = \frac{v_o}{v_i} = \frac{1 - g_m R_G}{1 + R_G/R_L} = -20.0\,V/V$$

$$A_v = \frac{v_o}{v_i} = \frac{1 - g_m R_G}{1 + R_G/(r_o \parallel R_L)} = -18.18\,V/V$$

Example

Find the voltage gain, input and output resistances of the amplifier circuit shown below. Take $V_t = 0.5\ V, V_A = 50\ V$ and $k = 2\ mA/V^2$.





$$v_i = (10\ mV)\sin \omega t$$

$$v_o = (-20)(10\ mV)\sin \omega t$$

Peak voltage at the output (Drain)

$$v_D = 1 \pm 0.2\ (peak)$$

$$v_G = V_G + v_i = 1 + v_i$$

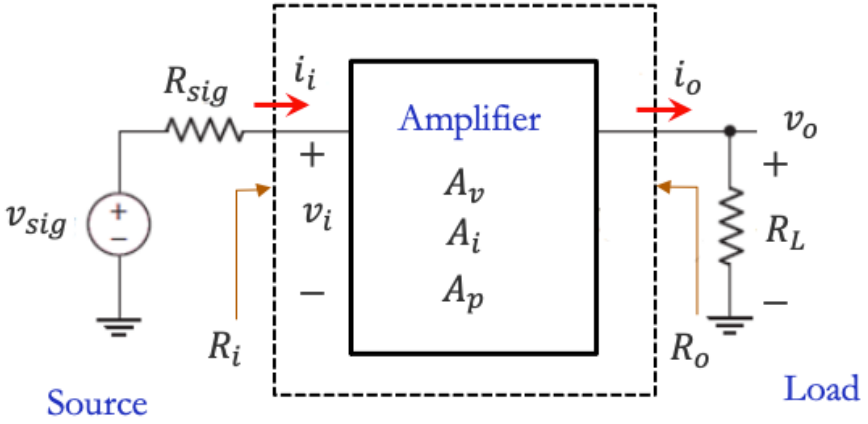
$$v_D = V_D + v_o = 1 + v_o$$

To stay in saturation $\quad v_D \geq v_G - V_t = 0.5\ V$

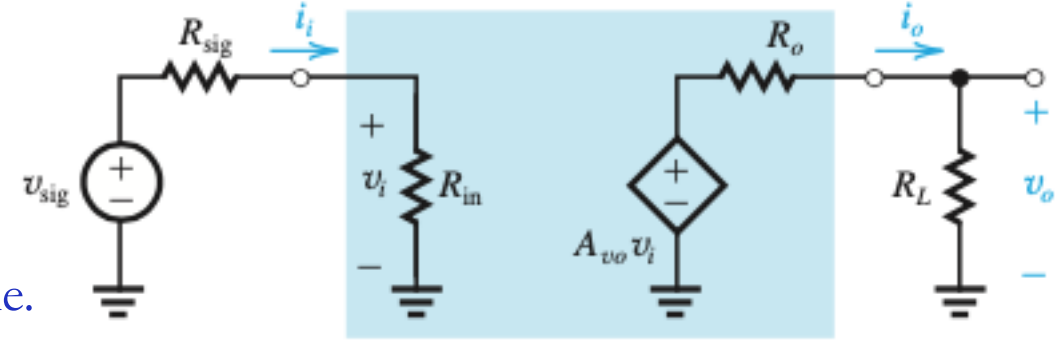
Maximum swing $\quad v_i \leq 25\ mV\ (peak)$

Input and Output Resistances

A good amplifier design takes care of energy coupling to and from the amplifier. This is satisfied by an appropriate design of the input and output resistances.



The equivalent circuit of any voltage amplifier.

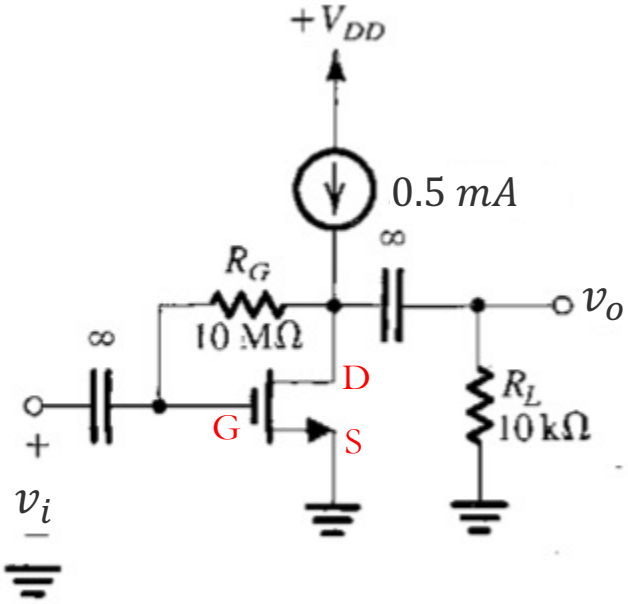


R_i Should be as high as possible.

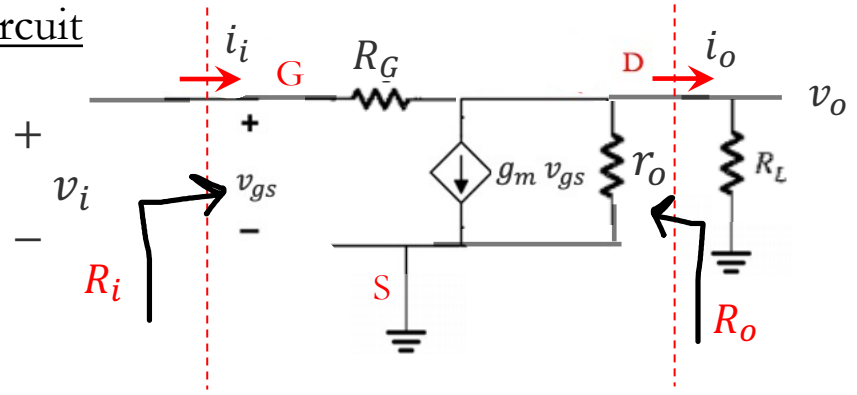
R_o Should be as low as possible.

Example

Find the voltage gain, input and output resistances of the amplifier circuit shown below. Take $V_t = 0.5\ V, V_A = 50\ V$ and $k = 2\ mA/V^2$.



AC circuit



To find the equivalent resistance, independent sources need to be properly removed.

Amplifier input resistance $R_i = \dfrac{v_i}{i_i}$

Amplifier output resistance $R_o = \dfrac{v_o}{i_o}$

Find R_i $\quad i_i = \dfrac{v_i - v_o}{R_G} \Rightarrow \dfrac{i_i}{v_i} = \dfrac{1 - v_o/v_i}{R_G}$

$$R_i = \frac{v_i}{i_i} = \frac{R_G}{1 - A_v} = 0.52\ M\Omega$$

Find R_o $\quad R_o = R_G \parallel r_o \cong 100\ k\Omega$

Q & A

Nano Photonics and Plasmonics Research

Send me an email if
you have any
questions.